

Mortality, Respiratory (ICD-9 code 460-519) - adjusted daily metric

Based on the coefficients estimated from the two-pollutant model in the 86 urban areas using daily 1-hour max metric, the coefficients were adjusted for daily 8-hour max metric using a ratio of 1.14 (Anderson and Bell 2010, Table 2).

G.2.1.7 Levy et al. (2005)

The authors conducted an empiric Bayes metaregression to estimate the ozone effect on mortality, and to assess whether this effect varies as a function of hypothesized confounders or effect modifiers. They gathered 71 time-series studies relating ozone to all-cause mortality, and they selected 48 estimates from 28 studies for the metaregression. Metaregression covariates included the relationship between ozone concentrations and concentrations of other air pollutants, proxies for personal exposure-ambient concentration relationships, and the statistical methods used in the studies. For the metaregression, they applied a hierarchical linear model with known level-1 variances. The authors estimated a grand mean of a 0.21% increase (95% confidence interval = 0.16-0.26%) in mortality per 10-microg/m increase of 1-hour maximum ozone (0.41% increase per 10 ppb) without controlling for other air pollutants. In the metaregression, air-conditioning prevalence and lag time were the strongest predictors of between-study variability. Air pollution covariates yielded inconsistent findings in regression models, although correlation analyses indicated a potential influence of summertime PM_{2.5}.

All-Cause Mortality

Levy et al. (2005, Table 1) reported a 0.43% change in all-cause mortality with a 95% confidence interval of 0.29% to 0.56% associated with a 10 µg/m³ change in ozone. We converted µg/m³ to ppb with an assumed relationship of 1.96 µg/m³ per 1.0 ppb.

G.2.1.8 Moolgavkar et al. (1995)

Moolgavkar et al. (1995) examined the relationship between daily non-accidental mortality and air pollution levels in Philadelphia, Pennsylvania from 1973 to 1988. They examined ozone, TSP, and SO₂ in a three-pollutant model, and found a significant relationship for ozone and SO₂; TSP was not significant. In season-specific models, ozone was significantly associated with mortality only in the summer months.

Non-Accidental Mortality

The health impact function for ozone is based on the full-year three-pollutant model reported in Table 5 (Moolgavkar et al., 1995, p. 482). The coefficient and standard error are based on the relative risk (1.063) and 95% confidence interval (1.018-1.108) associated with a 100 ppb increase in daily average ozone.

G.2.1.9 Samet et al. (1997)

Samet et al. (1997) examined the relationship between daily non-accidental mortality and air pollution levels in Philadelphia, Pennsylvania from 1974 to 1988. They

examined ozone, TSP, SO₂, NO₂, and CO in a Poisson regression model. In single pollutant models, ozone, SO₂, TSP, and CO were significantly associated with mortality. In a five-pollutant model, they found a positive statistically significant relationship for each pollutant except NO₂.

Non-Accidental Mortality

The health impact function for ozone is based on the five-pollutant model (ozone, CO, NO₂, SO₂ and TSP) reported in Table 9 (Samet et al., 1997, p. 20). The ozone coefficient and standard error are based on the percent increase (1.91) and t-statistic (3) associated with a 20.219 ppb increase in two-day average ozone.

G.2.1.10 Schwartz (2005)

The author used the case-crossover approach, where the control for each person is the same person on a day near in time, when he or she did not die. This method controls for season and individual risk factors by matching. One can also choose the control day to have the same temperature as the event day. The author applied this approach to a study of more than 1 million deaths in 14 U.S. cities. He found that, with matching on temperature, a 10-ppb increase in maximum hourly ozone concentrations was associated with a 0.23% (95% confidence interval [CI] 0.01%, 0.44%) increase in the risk of dying. This finding was indistinguishable from the risk when only matching on season and controlling for temperature with regression splines (0.19%; 95% CI 0.03%, 0.35%). Control for suspended particulate matter with an aerodynamic diameter of 10 µm or less (PM₁₀) did not change this risk. However, the association was restricted to the warm months (0.37% increase; 95% CI 0.11%, 0.62%), with no effect in the cold months. The author concluded that the association between ozone and mortality risk is unlikely to be caused by confounding by temperature.

Non-Accidental Mortality

Assuming a 10 ppb change in the daily 1-hour maximum, Schwartz (2005, Table 2) reported a 0.37% change in non-accidental mortality with a 95% confidence interval of 0.11% to 0.62%.

G.2.1.11 Smith et al. (2009)

Smith et al. (2009) analyzed the relationship between daily mortality and ambient ozone concentrations through re-examination of evidence using the National Morbidity, Mortality, and Air Pollution Study (NMMAPS), which collected daily data on mortality, meteorology, and air pollutant concentrations for 100 U.S. cities from 1987-2000. The authors examined the sensitivity of city-specific ozone-mortality estimates to treatment of meteorology and co-pollutants, dependence on different ozone metrics, use of air conditioning, regional and spatial variability, and non-linear exposure-response relationships.

Non-Accidental Mortality

Assuming a 10 ppb change in daily 8-hour maximum ozone concentration, Smith et al. (2009) reported a 0.40 (0.22)% population-weighted change in non-accidental mortality in a model without PM₁₀, and 0.27 (0.22)% population-weighted change in non-accidental mortality in a model with PM₁₀.

G.2.1.12 Zanobetti and Schwartz (2008)

See full study description under Zanobetti and Schwartz (2008) in Appendix F, Section F.1.3.

Zanobetti and Schwartz (2008) examined the issue of “mortality displacement” (i.e., deaths are occurring in frail individuals and exposure is only moving the day of death to a day slightly earlier) in 48 U.S. cities during the warm season (i.e., June-August) for the years 1989-2000. The authors estimated a series of lagged specifications and found an effect size of 0.53% (95%C.I.: 0.28-0.77) for lag 0-3 days for all-cause mortality.

G.2.2 Hospital Admissions

Table G-12 summarizes the additional health impact functions used to estimate the relationship between ozone and hospital admissions. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table G-12. Additional Health Impact Functions for Ozone and Hospital Admissions

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
All Respiratory	Burnett et al.	2001	Toronto, CAN	0-1	PM _{2.5}	D8HourMax	0.008177	0.002377	Log-linear	Warm season. 8-hour max from 1-hour max.
All Respiratory	Katsouyanni et al.	2009	14 U.S. cities	65-99		D8HourMax	0.000637	0.000400	Log-linear	Summer, penalized splines, 8df
All Respiratory	Katsouyanni et al.	2009	14 U.S. cities	65-99		D8HourMax	0.000614	0.000406	Log-linear	Summer, natural splines, 8df
Chronic Lung	Moolgavkar et al.	1997	Minneapolis, MN	65-99	PM ₁₀ , CO	D24HourMean	0.002800	0.001769	Log-linear	
Chronic Lung	Moolgavkar et al.	1997	Minneapolis, MN	65-99	PM ₁₀ , CO	D8HourMax	0.001960	0.001238	Log-linear	All year. 8-hour max from 24-hour mean
Pneumonia	Moolgavkar et al.	1997	Minneapolis, MN	65-99	PM ₁₀ , SO ₂ , NO ₂	D24HourMean	0.003800	0.001088	Log-linear	
Pneumonia	Moolgavkar et al.	1997	Minneapolis, MN	65-99	PM ₁₀ , SO ₂ , NO ₂	D8HourMax	0.002660	0.000762	Log-linear	All year. 8-hour max from 24-hour mean

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Chronic Lung (less Asthma)	Schwartz	1994	Detroit, MI	65-99	PM ₁₀	D24HourMean	0.005523	0.002085	Log-linear	All year
Chronic Lung (less Asthma)	Schwartz	1994	Detroit, MI	65-99	PM ₁₀	D8HourMax	0.003424	0.001293	Log-linear	All year. 8-hour max from 24-hour mean.
Pneumonia	Schwartz	1994	Detroit, MI	65-99	PM ₁₀	D24HourMean	0.005210	0.001300	Log-linear	All year.
Pneumonia	Schwartz	1994	Minneapolis, MN	65-99	PM ₁₀	D24HourMean	0.003977	0.001865	Log-linear	All year.
Pneumonia	Schwartz	1994	Detroit, MI	65-99	PM ₁₀	D8HourMax	0.003230	0.000806	Log-linear	All year. 8-hour max from 24-hour mean
Pneumonia	Schwartz	1994	Minneapolis, MN	65-99	PM ₁₀	D8HourMax	0.002784	0.001305	Log-linear	All year. 8-hour max from 24-hour mean.
All Respiratory	Schwartz	1995	New Haven, CT	65-99	PM ₁₀	D24HourMean	0.002652	0.001398	Log-linear	Warm season
All Respiratory	Schwartz	1995	Tacoma, WA	65-99	PM ₁₀	D24HourMean	0.007147	0.002565	Log-linear	Warm season
All Respiratory	Schwartz	1995	New Haven, CT	65-99	PM ₁₀	D8HourMax	0.001777	0.000936	Log-linear	Warm season. 8-hour max from 24-hour mean
All Respiratory	Schwartz	1995	Tacoma, WA	65-99	PM ₁₀	D8HourMax	0.004931	0.001770	Log-linear	Warm season. 8-hour max from 24-hour mean

G.2.2.1 Burnett et al. (2001)

Burnett et al. (2001) studied the association between air pollution and acute respiratory hospital admissions (ICD codes 493, 466, 464.4, 480-486) in Toronto from 1980-1994, among children less than 2 years of age. They collected hourly concentrations of the gaseous pollutants, CO, NO₂, SO₂, and ozone. Daily measures of particulate matter were estimated for the May to August period of 1992-1994 using TSP, sulfates, and coefficient of haze data. The authors report a positive association between ozone in the May through August months and respiratory hospital admissions, for several single days after elevated ozone levels.